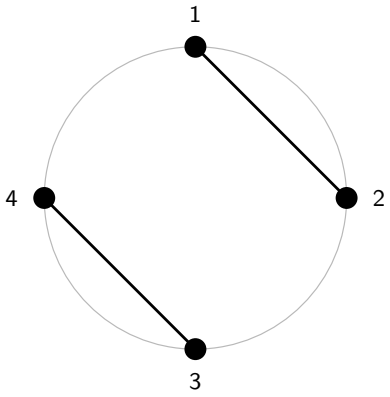
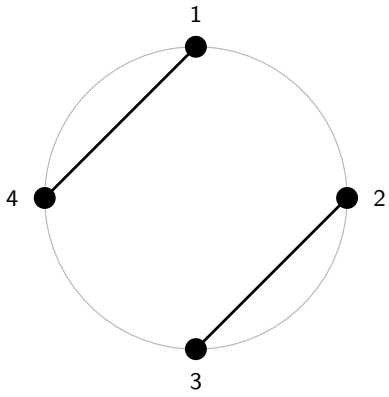
Week 12 / Catalan bijections / K–1

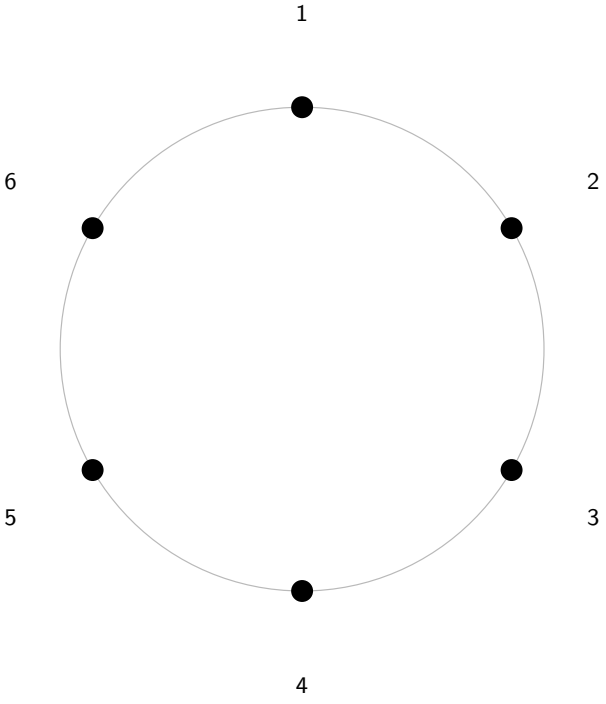
Use each dot once. Keep dots in place. Strings stay inside the circle without crossings. Pairings are different when a dot has a different partner. Use strings on large circles and a pencil on small circles.

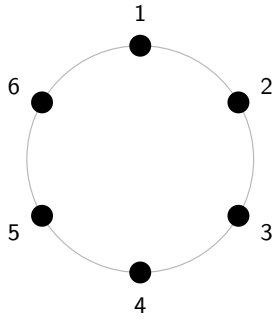
The two pairings of four dots:

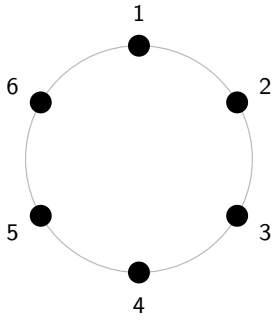


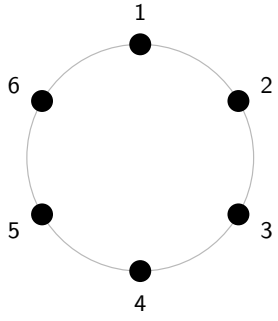


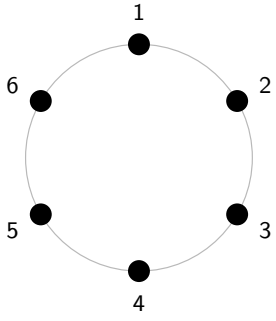
Problem 1: Find every way to pair six dots.

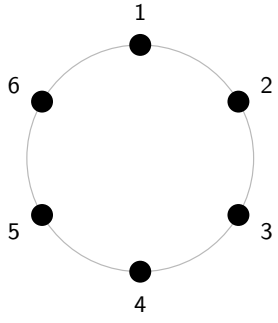


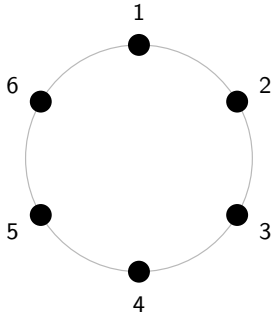




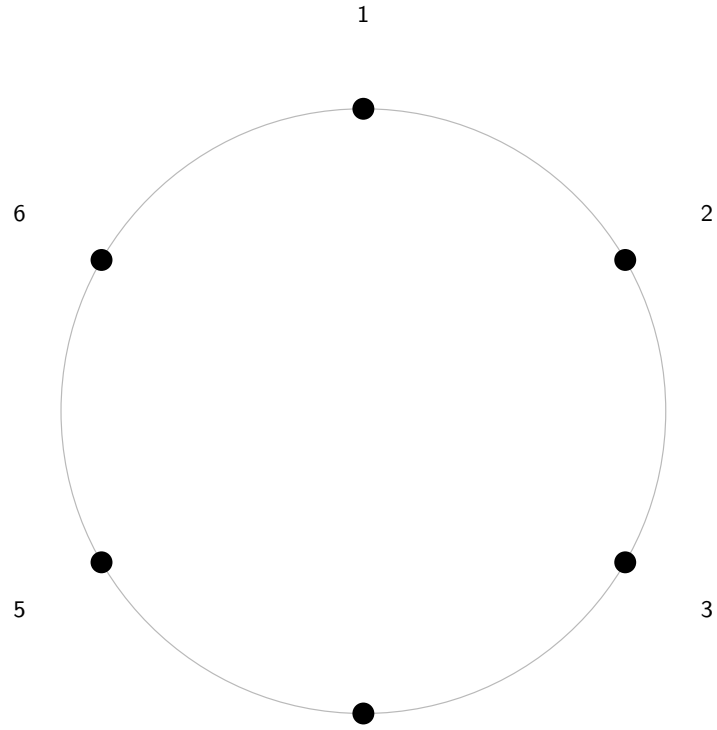


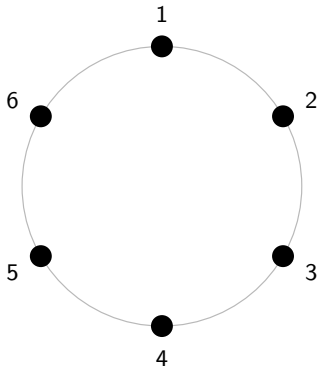


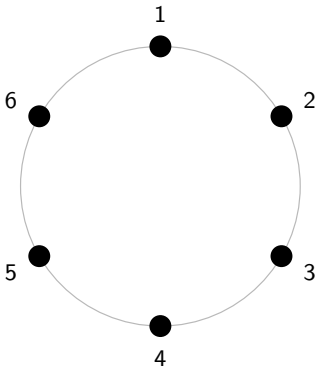


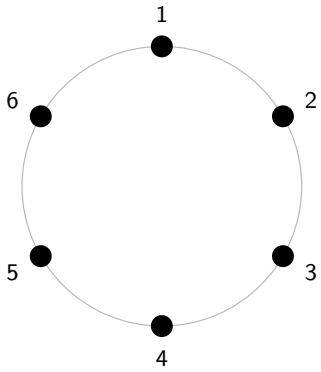


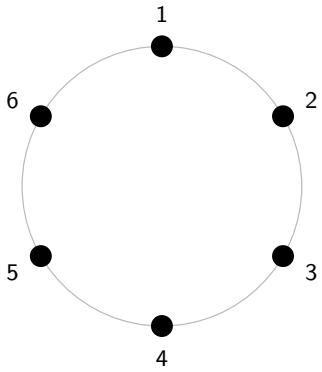
Problem 2: Check your six-dot collection. Show how you know that no pairing is missing.

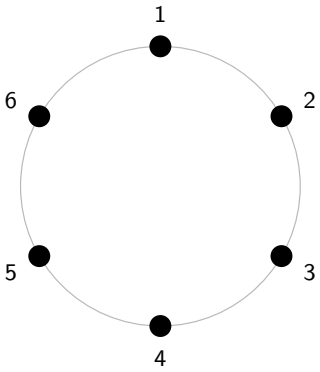


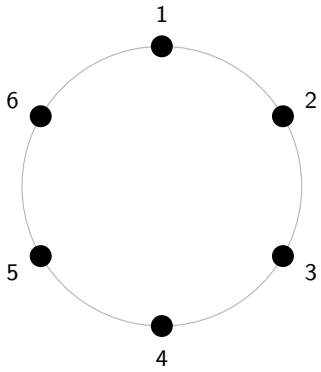




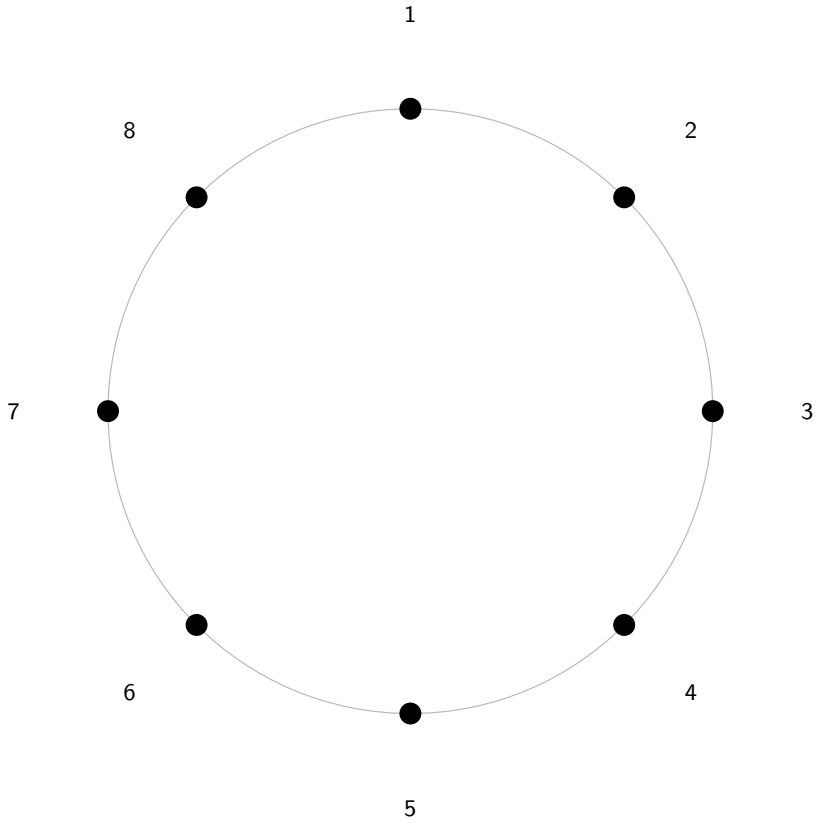


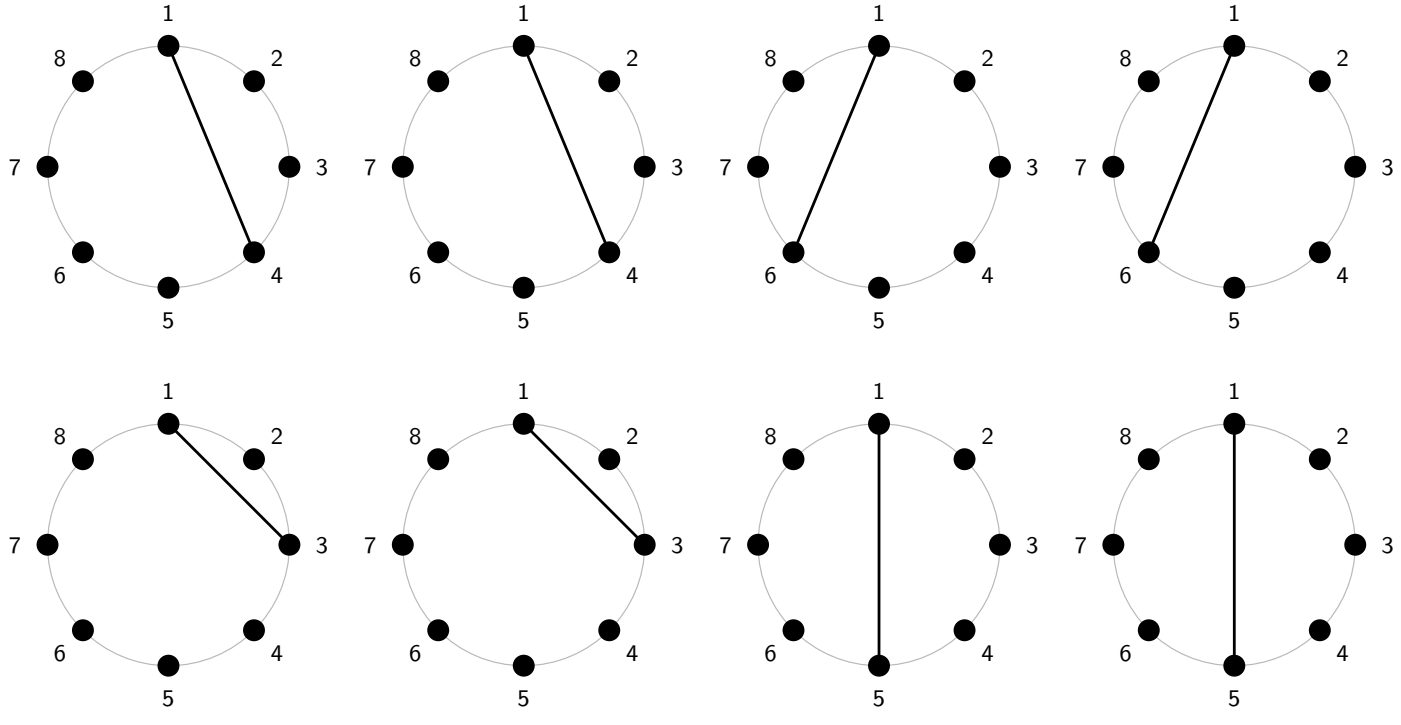




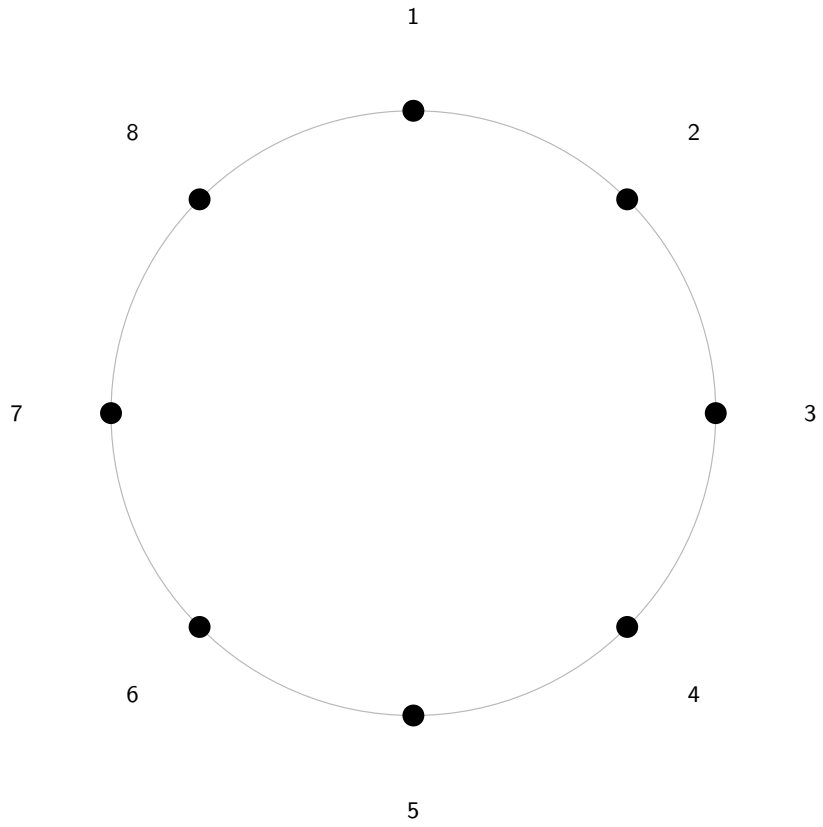


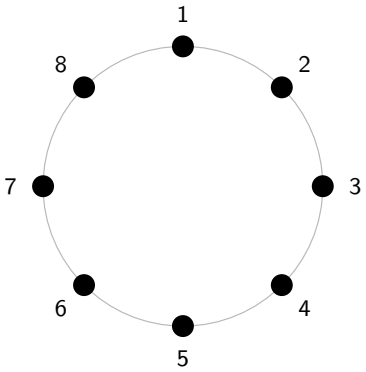
Problem 3: For each drawn pair, put a string on the large circle and leave it there. Find every way to finish the pairing, or show why you cannot.

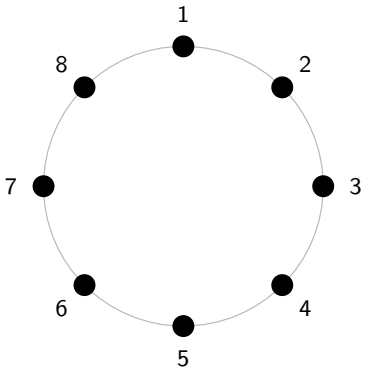


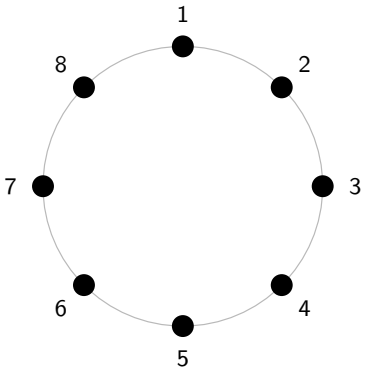


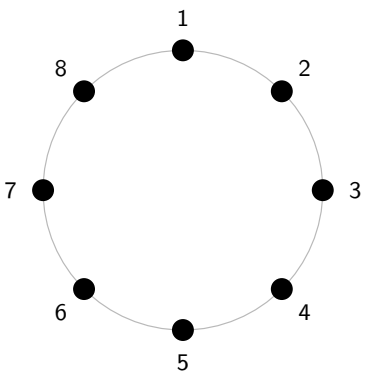
Problem 4: Pair eight dots in as many different ways as you can.

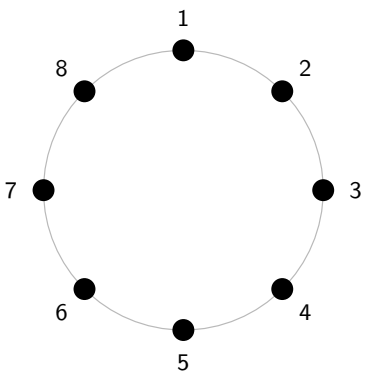


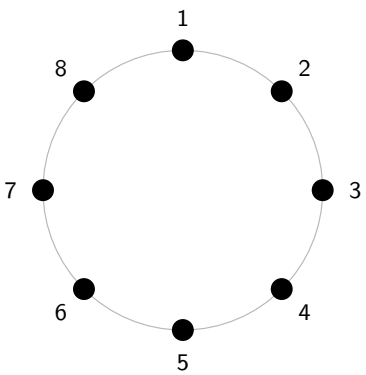




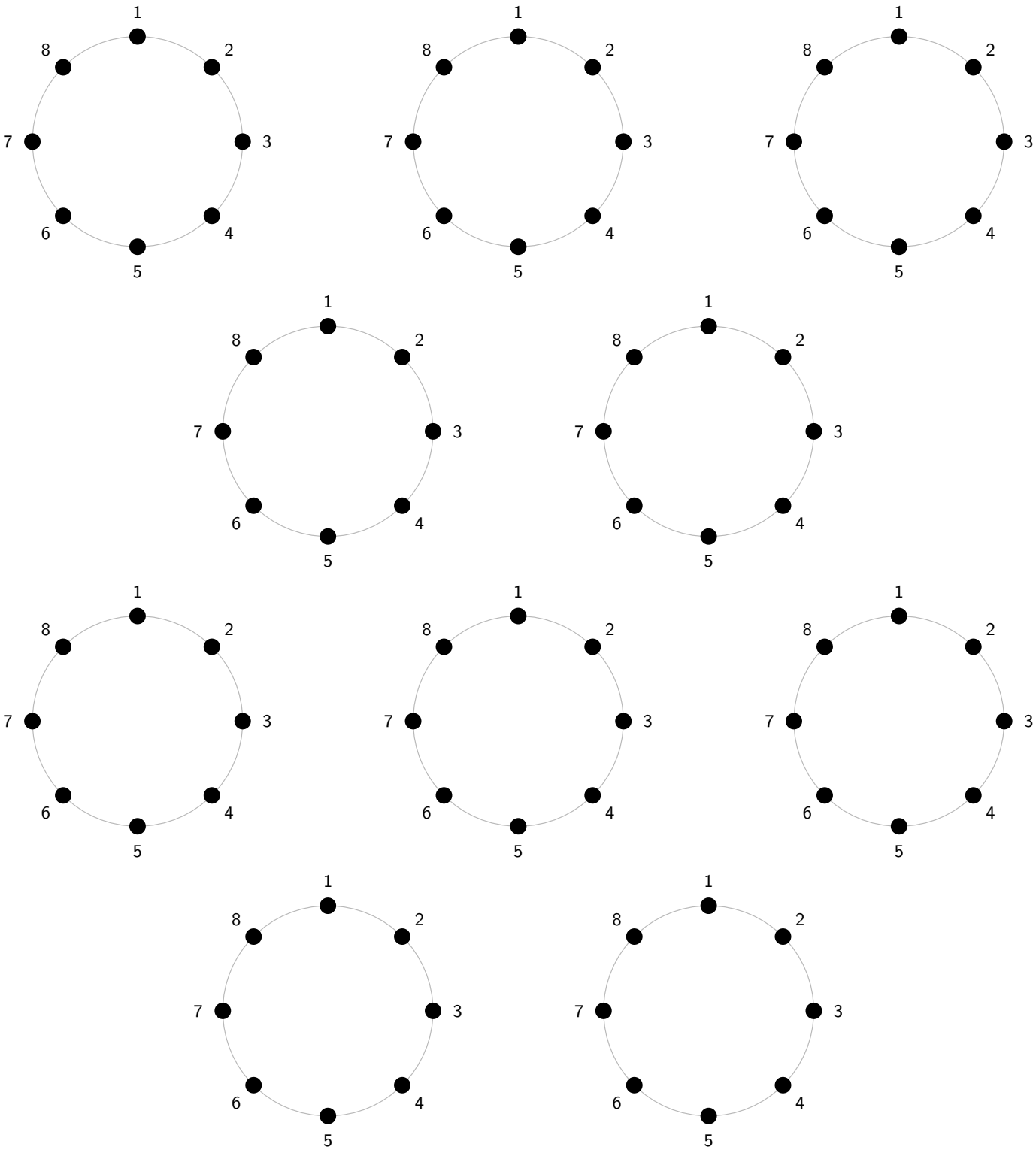




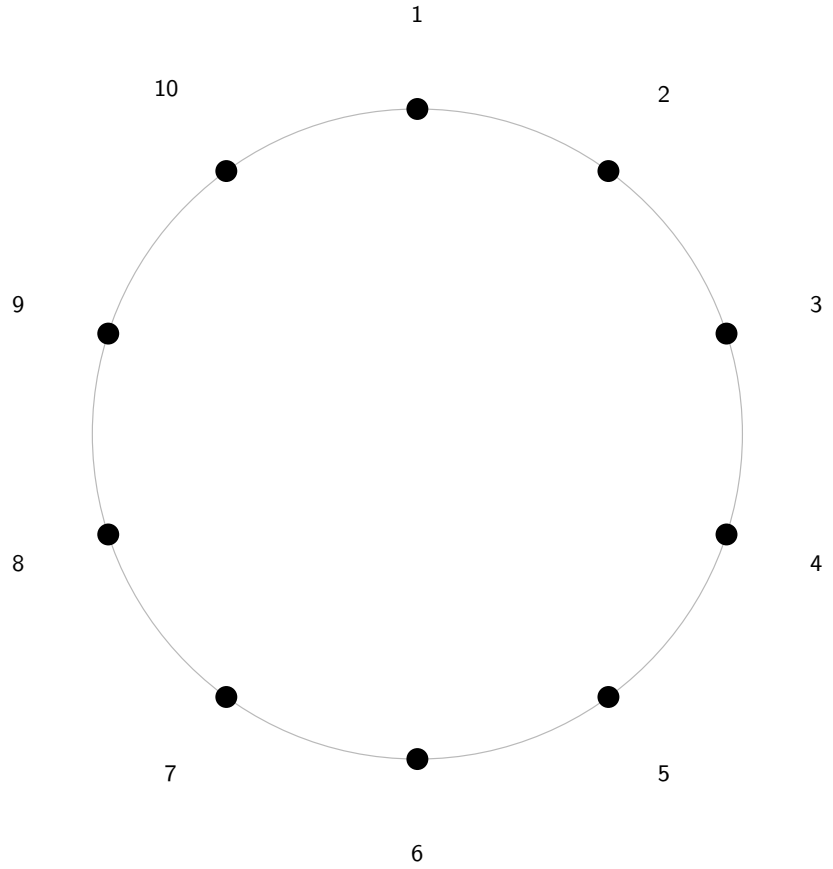


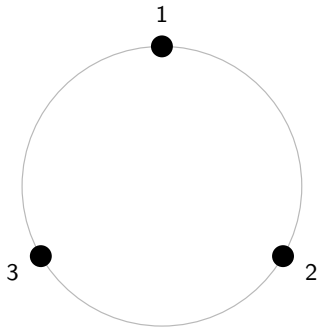


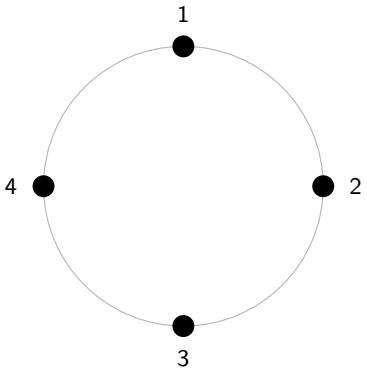
Problem 4 (continued)

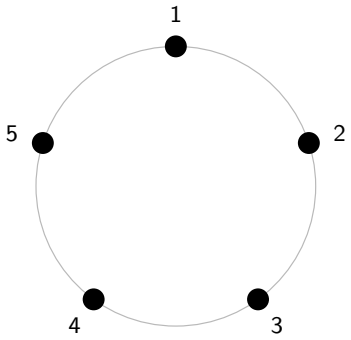


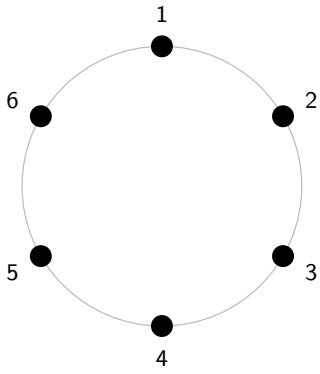
Problem 5: Pair all the dots in each circle if you can. If you cannot, show what gets in the way.

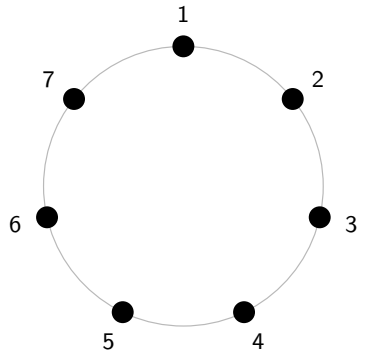












Problem 6: Can you pair eight dots without joining any two neighbors around the circle? Show why your answer is right.

